Programming Assignment Report

Design and Analysis of Algorithms

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Problem Name: Staircase

Problem Statement

In this challenge, you need to print a staircase of size **n** using the # symbol.
The staircase must be **right-aligned**, meaning each line contains spaces followed by # characters. The last line should contain no leading spaces.

Function Description

Complete the function **staircase** with the following parameter(s):

- **int n**: an integer representing the height and width of the staircase.

Return

No value should be returned. The function prints the staircase directly.

Input Format

- A single integer **n**, denoting the size of the staircase.

Output Format

Print a staircase of size **n** using # symbols and spaces.
All lines should be **right-aligned**, and the last line should not contain any leading spaces.

Example for **n = 6**

```
     #
    ##
   ###
  ####
 #####
```

######

Problem HackerRank Link:

https://www.hackerrank.com/challenges/staircase/problem

Github Submission Link:

https://github.com/Kishorevetrivel7/DAA-LAB

Solution Steps: Algorithm – Staircase

1. Start
2. Read the integer **n**
3. Use a loop from **1 to n**
4. For each row, print **(n – i) spaces**
5. Print **i number of # symbols**
6. Move to the next line
7. Repeat until the staircase is complete
8. Stop

Complete JAVA Code:

```
import java.io.*;
import java.util.*;

class Result {

  public static void staircase(int n) {
    for (int i = 1; i <= n; i++) {

      for (int j = 0; j < n - i; j++) {
        System.out.print(" ");
      }
```

```
            for (int k = 0; k < i; k++) {
                System.out.print("#");
            }

            System.out.println();
        }
    }

}

public class Solution {
    public static void main(String[] args) throws IOException {
        BufferedReader bufferedReader = new BufferedReader(new InputStreamReader(System.in));

        int n = Integer.parseInt(bufferedReader.readLine().trim());

        Result.staircase(n);

        bufferedReader.close();
    }
}
```

Observation/Solution:

1] The program imports the necessary Java libraries to handle input and output operations.

2] A **BufferedReader object** is created to read input from the keyboard.

3] The program reads an integer **n**, which represents the height and width of the staircase.

4] The **staircase(n)** function is called to generate and print the staircase pattern.

5] Inside the function, a **for loop** runs from **1 to n** to control the number of rows of the staircase.

6] For each row, another loop prints **(n – i) spaces** to ensure the staircase is right-aligned.

7] After printing spaces, a second loop prints **i number of # symbols** to form the step of the staircase.

8] Once the spaces and # symbols are printed, the program moves to the **next line** using System.out.println().

9] The loop continues until all **n rows of the staircase** are printed.

10] Finally, the program completes execution after displaying the full staircase pattern.